

MetalKB: Predicting Metal Binding Sites on Proteins with a Knowledge-Based Graph Framework

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Table S1: Comparison of the distance error distributions of MetalKB, PMM, and Metal3D. The deviation represents the mean distance $\pm$ standard deviation between the predicted and experimentally determined metal ion coordinates, and the median denotes the median of the distance error distribution. The results for MetalKB were calculated in this work, while those for PMM and Metal3D were taken from the PMM study^[1] .

Distance error distributions of Metal3D (E=1.7) on other metal datasets in Metal3D		
Metal	Deviation (Å)	Median (Å)
Mn	0.874 $\pm$ 1.088	0.502
Fe ⁺²	0.523 $\pm$ 0.737	0.347
Fe ⁺³	0.535 $\pm$ 0.689	0.349
Co	0.669 $\pm$ 0.822	0.365
Ni	0.371 $\pm$ 0.267	0.304
Cu	0.362 $\pm$ 0.424	0.254
Zn	0.425 $\pm$ 0.884	0.174
Ca	0.314 $\pm$ 0.526	0.178
Mg	0.558 $\pm$ 0.624	0.365
Na	0.581 $\pm$ 0.824	0.249
K	0.407 $\pm$ 0.608	0.253
Distance error distributions of Metal3D (p=0.75) on other metal datasets in Metal3D		
Metal	Deviation (Å)	Median (Å)
Mn	0.992 $\pm$ 0.719	0.818
Fe ⁺²	0.738 $\pm$ 0.565	0.539
Fe ⁺³	0.939 $\pm$ 0.761	0.676
Co	0.937 $\pm$ 0.803	0.598
Ni	0.617 $\pm$ 0.435	0.524
Cu	0.647 $\pm$ 0.461	0.526
Zn	0.580 $\pm$ 0.554	0.390
Distance error distributions of PMM (p=0.75) on other metal datasets in Metal3D		
Metal	Deviation (Å)	Median (Å)
Mn	0.776 $\pm$ 1.165	0.328
Fe ⁺²	0.503 $\pm$ 0.771	0.283
Fe ⁺³	0.506 $\pm$ 0.733	0.321
Co	0.725 $\pm$ 0.868	0.385
Ni	0.327 $\pm$ 0.376	0.285
Cu	0.370 $\pm$ 0.403	0.269
Zn	0.428 $\pm$ 0.927	0.172

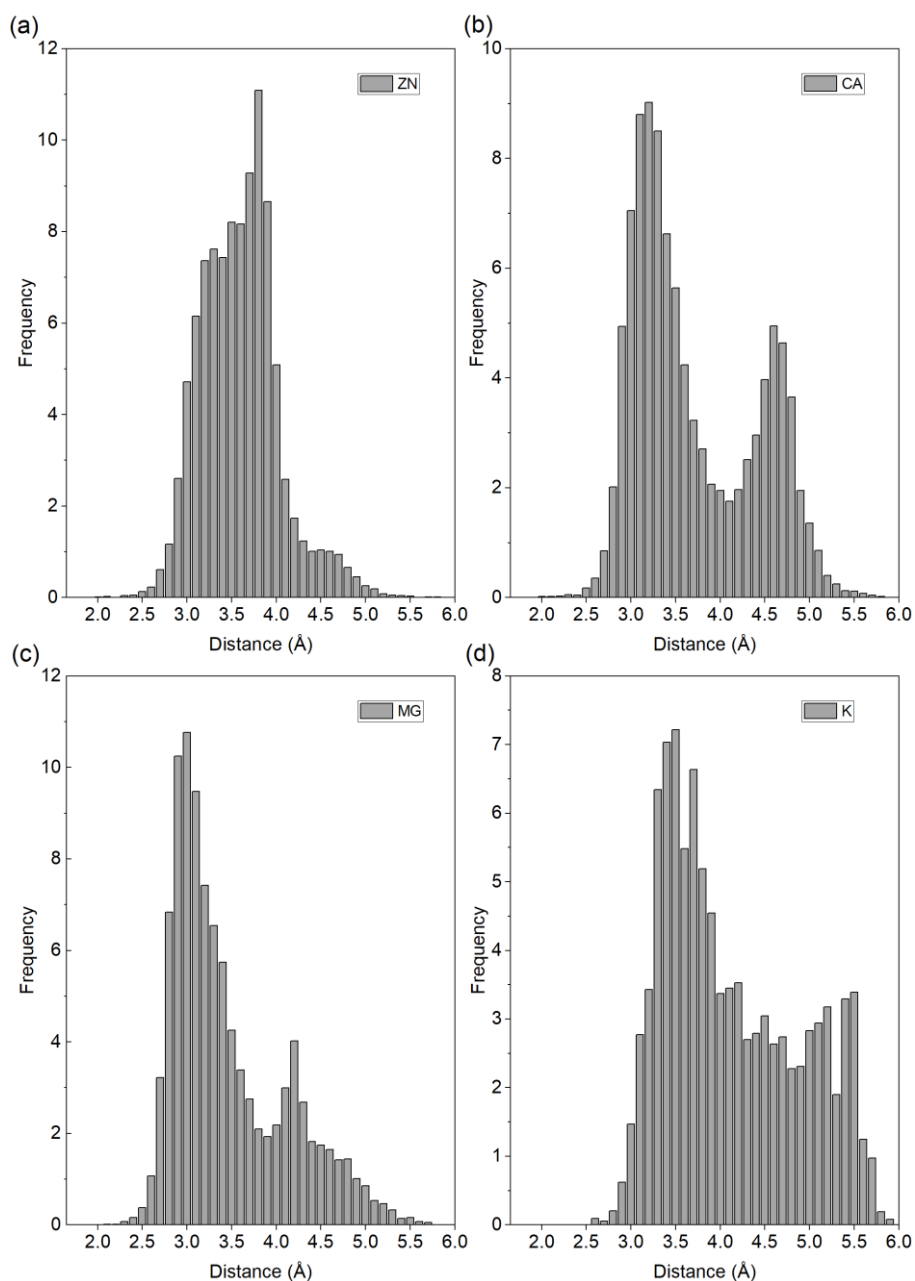


Figure S1: Statistical distributions of pairwise distances between donor atoms for different metal categories, including Zn (a), Ca (b), Mg (c), and K (d). Based on the observed distributions, the distance thresholds between donor atoms were set to 2.4-5.2 Å for Zn, 2.5-5.3 Å for Ca^{2+} , 2.5-5.3 Å for Mg^{2+} , and 2.9-5.8 Å for K^{+} , respectively. Because a virtual-atom representation was introduced for carboxylate oxygen atoms in the program, the distances between the two oxygen atoms of the same carboxylate group were not included in the statistics.

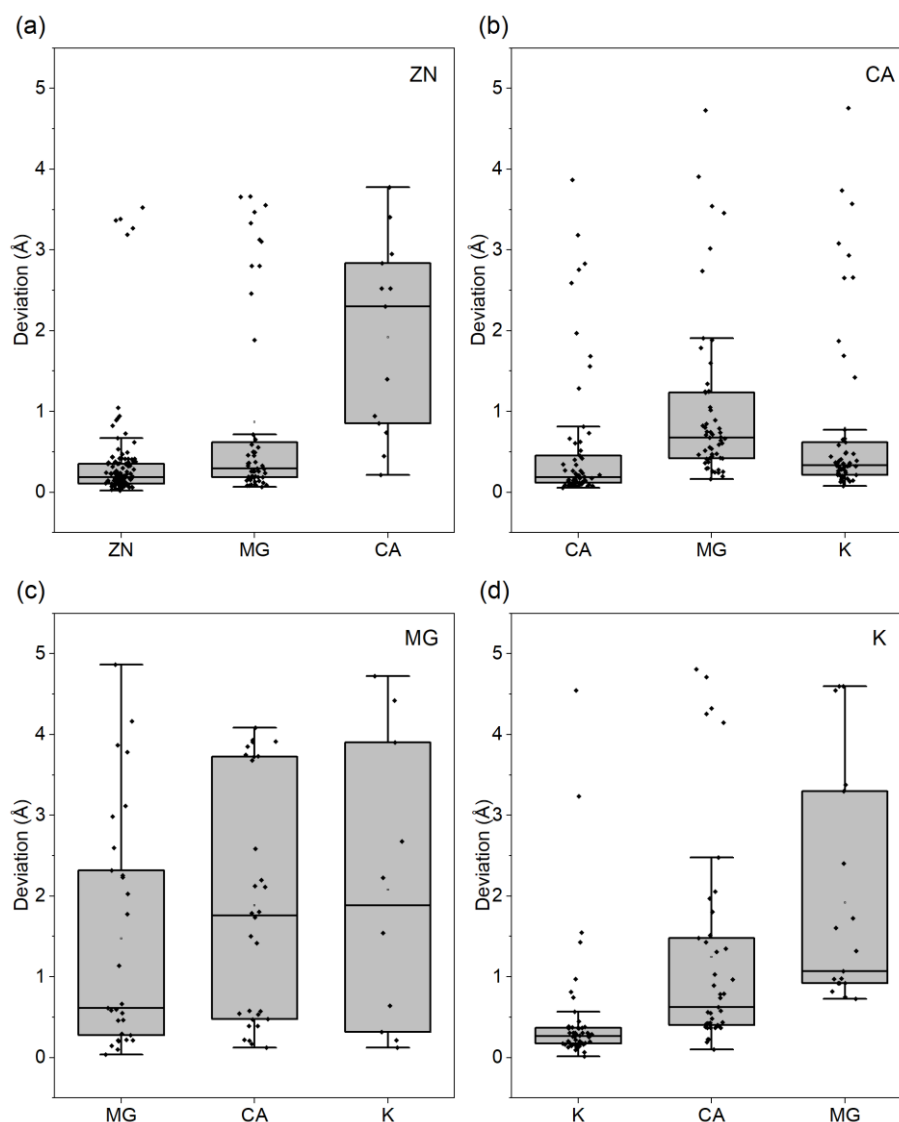


Figure S2: Distributions of spatial deviation distances between predicted metal positions (true positives, TPs) and experimentally annotated metal ions under different metal-specific statistical potentials. Each panel corresponds to binding sites of a given metal type, while the x-axis indicates the metal type associated with the statistical potential used for prediction. All predictions were obtained using a unified energy threshold of 1.7.

Cross-metal prediction analysis

To examine whether the energy scores obtained by MetalKB contain information relevant to distinguishing the metal type at a given binding site, we performed a cross-metal prediction analysis using datasets corresponding to different metal categories. The dataset used in this analysis is the same as that used for the results shown in Figure 4 of the main text. Specifically, it was drawn from the structural collection used to

derive the metal-specific statistical potentials, from which 100 metal-binding proteins were randomly selected for each metal type.

All predictions were generated using a unified energy threshold of 1.7. True positives (TP) were defined as predicted metal positions located within 5 Å of the experimentally annotated metal ion positions. Figure S2 shows the distributions of spatial deviation distances for the predictions, where only cases in which the number of predicted TPs was no less than 5% of the corresponding number of true sites are displayed.

As shown in Figure S2, the statistical potentials exhibit a certain degree of metal-type specificity. When the statistical potential corresponding to the true metal type is applied, the resulting spatial deviation distributions are generally more concentrated and show smaller overall deviations. This indicates that the statistical potentials impose more appropriate geometric constraints on binding sites that are chemically and structurally compatible with the corresponding metal types.

At the same time, when statistical potentials corresponding to metal types different from the experimentally annotated ones are applied, a fraction of predictions still satisfy the TP distance criterion. This phenomenon is particularly pronounced for metal types with similar coordination geometries and donor atom compositions. This behavior reflects, on the one hand, the intrinsic similarity of some metal coordination environments in protein structures, and on the other hand, the characteristics of the current statistical potential framework, which is primarily based on distance-dependent and geometric preferences rather than detailed electronic structure effects.

Overall, these results indicate that reliance on the current energy function alone is insufficient to achieve precise discrimination among metal types. It should be noted that the primary design goal of MetalKB is the identification and spatial localization of metal-binding sites, rather than precise metal type classification. Achieving fine-grained discrimination among specific metal types will require further investigation based on the current framework.

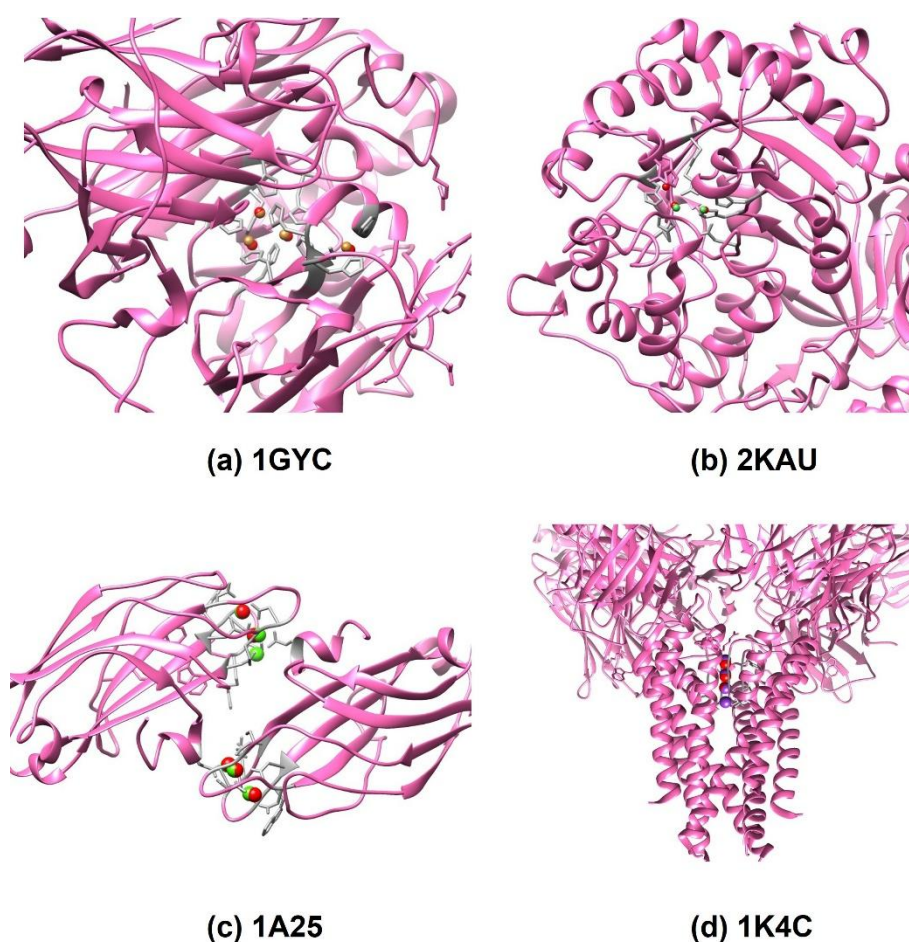


Figure S3: Representative examples of MetalKB predictions for non-zinc metal-binding proteins. (a) The multi-copper oxidase laccase (PDB ID: 1GYC), showing the trinuclear copper cluster in the catalytic center. (b) The nickel-dependent urease from *Klebsiella aerogenes* (PDB ID: 2KAU), featuring a dinuclear Ni^{2+} active site. (c) The Ca^{2+} -bound C2 domain of protein kinase C (PDB ID: 1A25), showing multiple spatially adjacent Ca^{2+} ions. (d) The potassium channel KcsA (PDB ID: 1K4C), showing four closely spaced K^{+} ions within the selectivity filter. In all panels, predicted metal ions are shown as red spheres, whereas experimentally determined metal ions are displayed as yellow (Cu), green (Ni), light green (Ca), and purple (K) spheres.

- [1] Zhang H; Zhong J; Gucwa M; Zhang Y; Ma H; Deng L; Mao L; Minor W; Wang N; Zheng H. PinMyMetal: a hybrid learning system to accurately model transition metal binding sites in macromolecules. *Nat Commun.* 2025, 16(1), 3043.